STRATIFYING THE BASE SO THAT AN ETALE MORPHISM BECOMES FINITE ETALE

THE STACKS PROJECT AUTHORS

Editorial note. All delivered mathematical text is the author's own, copied from the pinned Stacks Project chapter file etale-cohomology.tex; no statement or proof step was rewritten, reordered or removed.

Editorial note. The proof is the author's complete argument for both cases and for the combination of the two stratifications.

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Editorial note. The checked chapter list for same-name collisions is the set of stacks chapters staged in this repository: chow, cohomology, divisors, etale-cohomology, flat, more-algebra, more-morphisms, morphisms, spaces-more-morphisms, varieties together with this card's chapter etale-cohomology.

Editorial note. The preamble is the chapter's own native preamble excerpt already used for this source: the Xy-pic setup (native preamble lines 14-18) and the native theorem declarations with the mathematical macros (native preamble lines 154-217).

Allowed standard prerequisites (editorial).

- Constructible locally closed subschemes and partitions of a quasi-compact quasi-separated scheme. Same chapter, the notion of a constructible locally closed subscheme and the existence of finite partitions into such subschemes. Supplies the strata of the partition produced in the proof.
- Finiteness of an etale morphism over a suitable locally closed stratum. Same chapter together with the chapter Morphisms, the description of an etale morphism as finite etale after restricting to a suitable locally closed subscheme, together with the separated and quasi-finite reductions. Supplies the construction of the partition stratum by stratum.

Source reference. Original lemma 07RZ: sources/stacks-etale-cohomology/pinned-original-source.tex, source lines 12044–12044. Navigation pointer invoked at proof line 12044; the counted span contains the pointer only and the cited result is not reproduced inside it, so the pointer is registered as a non-load-bearing navigation stub in the established form for Stacks material. No bibliography citation is involved.

Source reference. Original lemma 01W6: sources/stacks-etale-cohomology/pinned-original-source.tex, source lines 12055–12055. Navigation pointer invoked at proof line 12055; the counted span contains the pointer only and the cited result is not reproduced inside it, so the pointer is registered as a non-load-bearing navigation stub in the established form for Stacks material. No bibliography citation is involved.

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Source reference. Original lemma 01KP: sources/stacks-etale-cohomology/pinned-original-source.tex, source lines 12058–12058. Navigation pointer invoked at proof line 12058; the counted span contains the pointer only and the cited result is not reproduced inside it, so the pointer is registered as a non-load-bearing navigation stub in the established form for Stacks material. No bibliography citation is involved.

Lemma 0.1. *Let $U \to X$ be an étale morphism of quasi-compact and quasi-separated schemes (for example an étale morphism of Noetherian schemes). Then there exists a partition $X = \coprod_i X_i$ by constructible locally closed subschemes such that $X_i \times_X U \to X_i$ is finite étale for all i.*

Proof. If $U \to X$ is separated, then this is More on Morphisms, Lemma 07RZ. In general, we may assume X is affine. Choose a finite affine open covering $U = \bigcup U_j$. Apply the previous case to all the morphisms $U_j \to X$ and $U_j \cap U_{j'} \to X$ and choose a common refinement $X = \coprod X_i$ of the resulting partitions. After refining the partition further we may assume X_i affine as well. Fix i and set $V = U \times_X X_i$. The morphisms $V_j = U_j \times_X X_i \to X_i$ and $V_{jj'} = (U_j \cap U_{j'}) \times_X X_i \to X_i$ are finite étale. Hence V_j and $V_{jj'}$ are affine schemes and $V_{jj'} \subset V_j$ is closed as well as open (since $V_{jj'} \to X_i$ is proper, so Morphisms, Lemma 01W6 applies). Then $V = \bigcup V_j$ is separated because $\mathcal{O}(V_j) \to \mathcal{O}(V_{jj'})$ is surjective, see Schemes, Lemma 01KP. Thus the previous case applies to $V \to X_i$ and we can further refine the partition if needed (it actually isn't but we don't need this). □